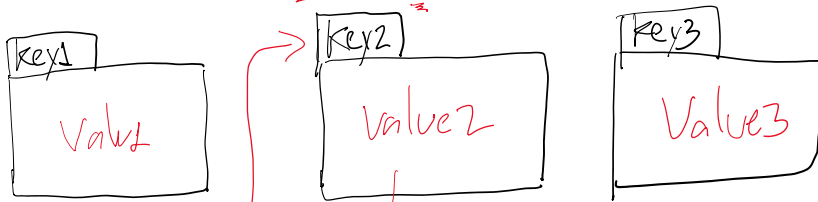


Dictionary

- Data structure ที่เก็บข้อมูลแบบไม่เรียงลำดับ
- ประกอบด้วยคู่ (pair) key กับ value
- Syntax:
$$\text{dict} = \{ \text{key1: value1}, \text{key2: value2}, \text{key3: value3} \}$$

#1 pair #2 pair #3 pair

* Access: 6 วิธี ใน value หรือ key
 $\text{dict}[\text{key1}] \rightarrow \text{value1}$



access $\rightarrow$ key2 $\rightarrow$ value2

key: วัตถุแบบ immutable object $\rightarrow$ string, int, float, tuple

$\uparrow$ key
คือ string

value: วัตถุแบบ mutable $\rightarrow$ dict, list, string, ...

Access value หรือ key

Syntax: $\text{dict}[\text{key}] \rightarrow \text{value}$

Ex) $\text{my_dict} = \{ \text{'A': 1}, \text{'B': 2}, \text{'C': 3} \}$

$\text{print}(\text{my_dict}[\text{'B'}]) \rightarrow 2$

$x = \text{my_dict}[\text{'A'}] + 2 * \text{my_dict}[\text{'C'}]$

$\text{print}(x) \rightarrow 7$
 $1 + 2 * 3 \rightarrow 7$

print(x) → 7

Ex2 names = {103: "Alice", 105: "Bob", 106: "Chris"}

print(names[105]) → Bob

print(names[103].lower() + names[106].upper())
"Alice".lower() + "Chris".upper()
aliceCHRIS

Update / Modify value

Syntax: dict[key] = new_value

grades = {"Alice": 'A', "Bob": 'B', "Ann": 'D'}

print(grades["Alice"]) → A

grades["Alice"] = 'C' → "Alice": 'C'

print(grades["Alice"]) → C

* Missing or wrong key in dict

print(grades["Chris"])

grades["Chris"] → Error!

• in operator

↳ check if key is in dict or not

"Alice" in grades → True

"Chris" in grades → False

• len() function

↳ number of items

len(grades) = 3

len()

↳ တစ်ခုတည်းပဲ

Ex

len(grades) → 3

သိသေး 3 ခု

သိသေး ၃ ခု ဖြစ်သော dict

Syntax: dict [new-key] = new-value

သိသေး new-key: new-value ဖြစ်သည်

Ex1 ဒီ dict ကို အကယ်၍ empty ({})

grades = {} ← empty dict

grades["Anna"] = 'B'

grades["Mathew"] = 'C'

grades["Bob"] = 'A'

သိသေး dict ကို

print(grades) → {'Anna': 'B', 'Mathew': 'C', 'Bob': 'A'}

Ex2 ဘယ်လို သိသေး dict ကို list

names = ["Alice", "Anna", "Bob"]

scores = [70, 80, 77]

students = {} → name: score → {'Alice': 70, 'Anna': 80, 'Bob': 77}

for i in range(len(names)): Brute-force 'Alice' 70

students[names[i]] = scores[i] → students[names[0]] = scores[0]

→ students[names[1]] = scores[1]

→ students[names[2]] = scores[2]

print(students)

↓
{'Alice': 70, 'Anna': 80, 'Bob': 77}

ဘယ်လို dictionary

1. for loop ဝဲ ဘယ်လို key

↳ for loop of only key

↳ also works for loop of only key's

EX my_dict = {'A': 1, 'B': 2, 'C': 3}

for k in my_dict:

print(k)

print(my_dict[k])

A
1
B
2
C
3

Useful dictionary Methods

• keys() → returns all keys as 'list'

• values() → returns all values as 'list'

• items() → returns all key-value pair as 'list' of tuple

* If
you
want
loop!

EX1

my_dict = {'A': 1, 'B': 2, 'C': 3}

for v in my_dict.values():

print(v)

for k, v in my_dict.items():

print(k, v)

1
2
3
A 1
B 2
C 3

• get(key, default)

• get(key , default)

→ If value is given key is
if key is not in dict then default value

```
def my_get( my_dict, key, default):  
    if key in my_dict:  
        return my_dict[key] ←  
    else:  
        return default
```

• Use case: count new dictionary

- Input: list of words
→ ['apple', 'apple', 'banana', 'orange', 'apple']

→ count how many words are there

→ list of words: apple, banana, orange

#1 fruits = ['apple', 'apple', 'banana', 'orange']

apple_n = 0

banana_n = 0

orange_n = 0

for fruit in fruits:

if fruit == 'apple':

apple_n += 1

if fruit == 'orange':

orange_n += 1

if fruit == 'banana':

banana_n += 1

idea: dict an count how many

key is there

value is count

count_f = {'apple': 0, 'banana': 0, 'orange': 0}

for fruit in fruits:

count_f[fruit] += 1

count_f[fruit]
= count_f[fruit] + 1

↑
update count
of fruit

Ex in list version

sentence = "AA BB AA This is cat meow meow meow"

words_c = {}

→ default is split

sentence = "AA BB AA This is cat meow meow meow"

words_c = {}

words = sentence.split() *default is split*
list of string
['AA', 'BB', 'AA', 'This', 'is', 'cat', 'meow', 'meow', 'meow']

for word in words:

if word in words_c:

words_c[word] = words_c[word] + 1

else:

words_c[word] = 0 + 1

error in word key in words_c

words_c[word]
= words_c.get(word, 0) + 1

Nested Dictionary

Dict rows dict (JSON)

Users = {"Alice": {"grade": "F", "age": 35},
"Bob": {"grade": "A", "age": 20},
"Anna": {"grade": "D", "age": 25}}

print(Users["Anna"]) → {'grade': 'D', 'age': 25}

print(Users["Anna"]["grade"]) → D
{'grade': 'D', 'age': 25}

Dict rows list (value as dict in list)

my_dict = {"Alice": [10, 15, 25],
"Bob": [20, 30, 22]}

print(my_dict["Alice"]) → 10 15 25

print (my_dict ['Alice']) → [10, 15, 25]

print (my_dict ['Alice'] [0]) → 10

print (sum(my_dict ['Alice'])) → 50

my_dict ['Alice'].append (20)

print (my_dict) → { 'Alice': [10, 15, 25, 20],
 'Bob': [20, 30, 22] }